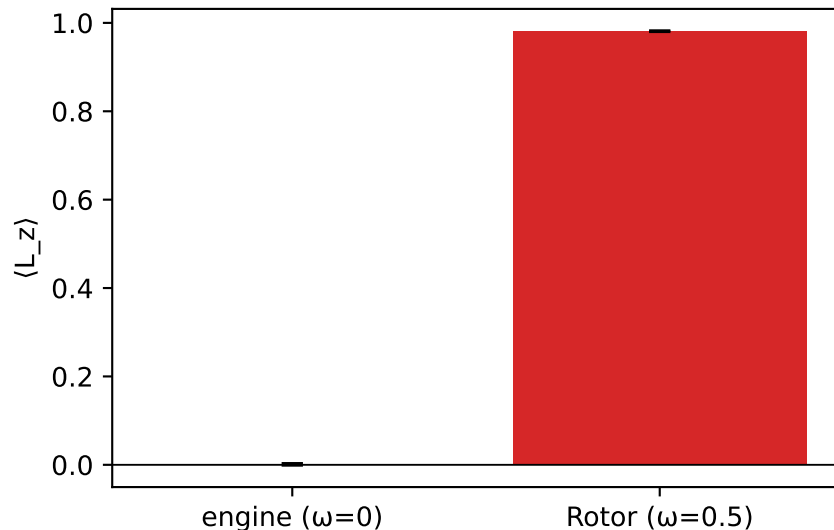
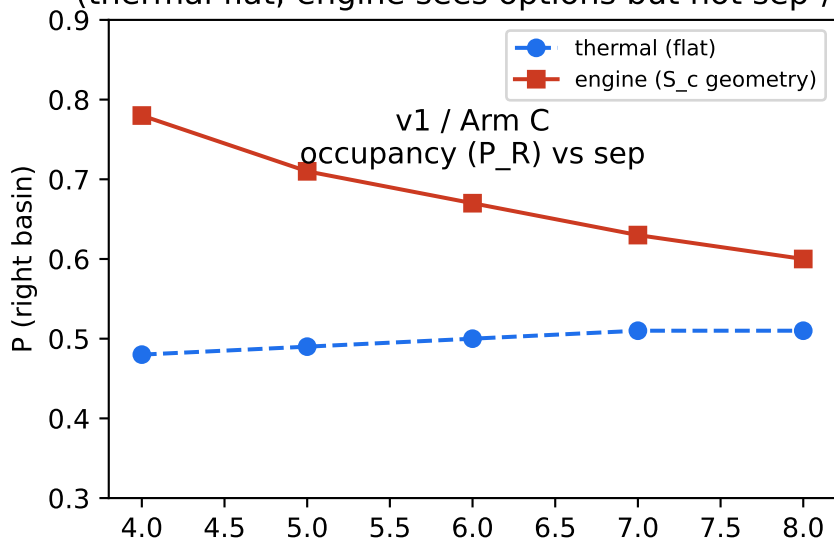


Fig 7 — Arc (honest): v3 re-confirms v2 thermodynamic equivalence in conservative 1D.

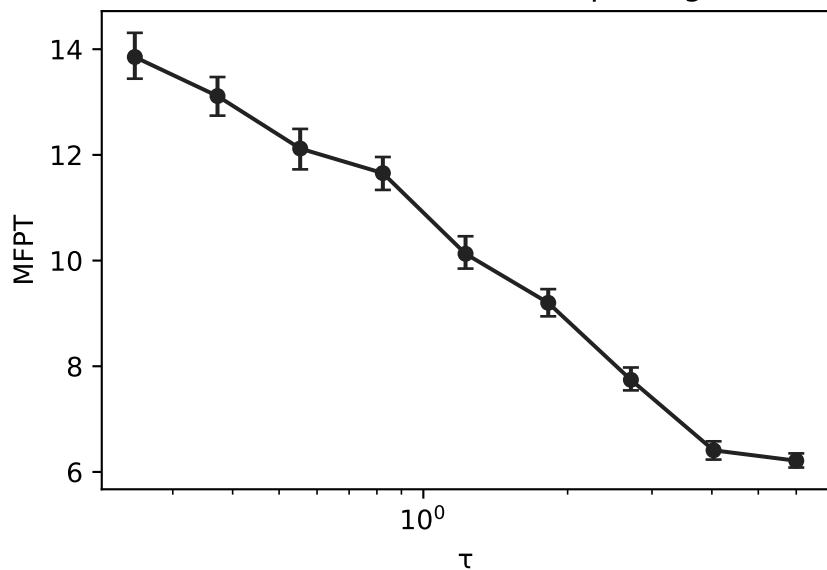
The negative (no unique  $\tau$ -fingerprint beyond  $U_{\text{eff}}$ ) is the result. Frontier = v4 non-conservative. Data: manifest (8000 traj/cell).

v2: current test — engine = 0 (conservative)  
positive control detects real NESS  
(this is the measurement)

v1: occupancy blind to L (detailed balance)  
(thermal flat; engine sees options but not  $\text{sep}^2/D$ )



v3: MFPT( $\tau$ ) dial  $\xrightarrow{\text{sep}}$  monotone, kinetic  
(the horizon moves first-passage)



v3 honest negative (this run)

- $\tau$  produces a visible FPT dial (real).
- But the dial *is* equilibrium relaxation in the  $\tau$ -dependent  $U_{\text{eff}}(\tau)$  — by definition once Arm A (conservative) is granted.
- $\alpha \approx 1$  (drift under force), not thermal diffusive  $\alpha=2$ .
- Zero current re-confirms thermodynamic equivalence (fig 5).

Conclusion: in conservative 1D the  $\tau$ -knob does not produce a fingerprint no static  $U$  can mimic. The frontier is v4 (non-conservative force, 2D+, curl  $\neq 0$ ).